

# Supporting Information — A Lineshape Correlation Coefficient for Robust Similarity of Two-Dimensional HSQC NMR Spectra

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Software (LCC and the three reference methods), both benchmark harnesses, all tabulated scores and the test suite are freely available under the MIT license at [https://github.com/deepnmr/spectra\\_similarity\\_toolbox](https://github.com/deepnmr/spectra_similarity_toolbox). LCC is implemented in `hsqc_lcc.py`.

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## S1 Setup and preprocessing

Two spectra  $a \in \{x, y\}$  are compared inside a common window  $\Omega = [\delta_{\text{lo}}^{F2}, \delta_{\text{hi}}^{F2}] \times [\delta_{\text{lo}}^{F1}, \delta_{\text{hi}}^{F1}]$  (direct dimension  $F2$ , e.g.  $^1\text{H}$ ; indirect dimension  $F1$ , e.g.  $^{15}\text{N}$  or  $^{13}\text{C}$ ). Each processed point  $p$  has shifts  $\delta_{F2}(p)$ ,  $\delta_{F1}(p)$  and intensity  $I_a(p)$ . With the default `clip` baseline, negative intensities are zeroed and every point is weighted by its local integration area so that spectra of different digital resolution integrate consistently:

$$w_a(p) = \max(I_a(p), 0) \cdot |\Delta\delta_{F1}(p)| \cdot |\Delta\delta_{F2}(p)|. \quad (\text{S1})$$

**Negative peaks (edited HSQC).** `clip` is correct for a non-edited spectrum, where negatives are baseline noise. In a **multiplicity-edited** HSQC, however, the sign encodes multiplicity ( $\text{CH}/\text{CH}_3$  positive,  $\text{CH}_2$  negative), not absence [2] — `clip` would silently delete every  $\text{CH}_2$  crosspeak. For

such data use `baseline="abs"`, which weights by  $|I_a(p)|$  (replacing  $\max(I_a, 0)$  in Eq. S1) and preserves the  $\text{CH}_2$  peaks; all methods and the CLI expose this option. The sparse  $^1\text{H}$ - $^{13}\text{C}$  peak-list benchmark already renders with `|intensity|` for exactly this reason (§S9).

This is exactly the area weighting of the bin method [1]. The mass is normalized to  $\sum_p w_a(p) = 1$ ; the final score is invariant to this scale, so normalization only guards against numerical overflow.

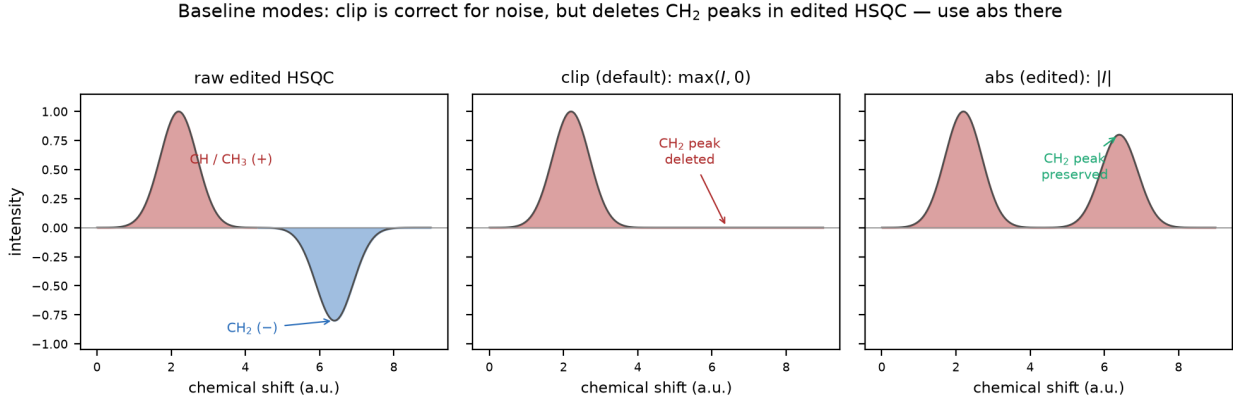


Figure 1: **Figure S1.** Baseline modes on a multiplicity-edited HSQC cross-section. **Left:** the raw signal, with a positive  $\text{CH}/\text{CH}_3$  peak and a **negative**  $\text{CH}_2$  peak (the sign encodes multiplicity, not absence). **Middle (clip, default):**  $\max(I, 0)$  zeroes negatives — correct when negatives are baseline noise, but it silently **deletes the  $\text{CH}_2$  crosspeak**. **Right (abs):**  $|I|$  preserves the  $\text{CH}_2$  peak. Use `clip` for a non-edited spectrum and `abs` for edited data; the sparse  $^1\text{H}$ - $^{13}\text{C}$  benchmark renders with `|intensity|` for exactly this reason (§S9).

## S2 Rendering to a shared grid

The window is discretized once into a fixed grid shared by **both** spectra:  $K$  bins along  $F1$  of width  $\Delta_1$  (default `step_f1` = 0.10 ppm) and  $L$  bins along  $F2$  of width  $\Delta_2$  (default `step_f2` = 0.01 ppm). Each spectrum is rendered by summing its weighted points into cells  $(k, l)$ :

$$R_a[k, l] = \sum_{p: \delta(p) \in \text{cell}(k, l)} w_a(p), \quad k = 1, \dots, K, \quad l = 1, \dots, L. \quad (\text{S2})$$

Rendering both spectra onto the *same* edges is what makes the two images directly comparable regardless of their native resolutions [3]. **Optional intensity compression:** to stop a single dominant amide from dominating the correlation, the rendered intensities may be raised to a power  $\rho$  (`--p`, default  $\rho = 1$ , i.e. off):  $R_a[k, l] \leftarrow R_a[k, l]^\rho$ . Both the unit-mass normalization and this optional compression are the standard scaling/normalization operations of quantitative NMR profiling [4,5].

## S3 Lineshape blur

The discrete image is convolved with a separable Gaussian whose width is the physical NMR linewidth *plus* the expected chemical-shift drift, one  $\sigma$  per axis:

$$\sigma_{F2} = \sqrt{\ell_{F2}^2 + d_{F2}^2}, \quad \sigma_{F1} = \sqrt{\ell_{F1}^2 + d_{F1}^2}, \quad (\text{S3})$$

Rendering onto shared edges makes two spectra comparable regardless of native resolution

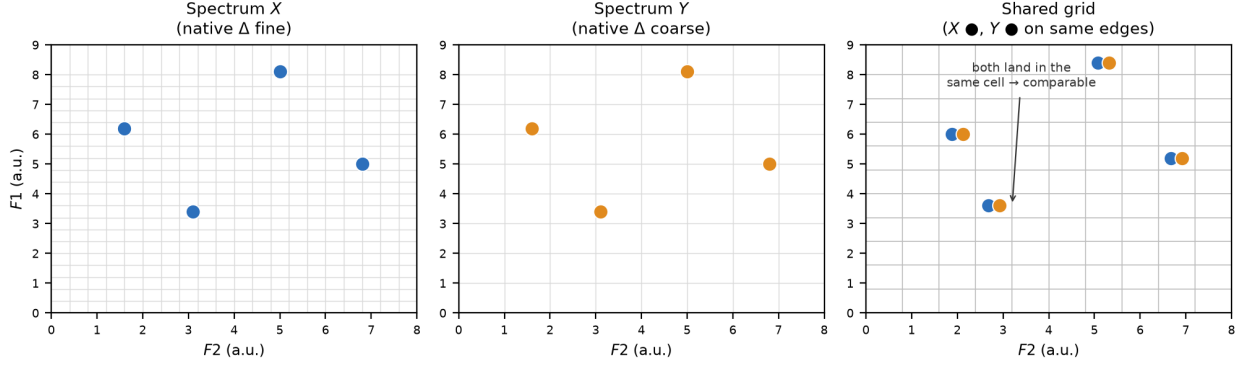


Figure 2: **Figure S2.** Rendering onto a shared grid. The **same** four crosspeaks are recorded by two instruments at different native digital resolutions (**left**, fine  $\Delta$ ; **middle**, coarse  $\Delta$ ). Histogramming both onto one **shared** grid with identical cell edges (**right**) places corresponding peaks in the **same cell**, so the two images become directly comparable regardless of their native resolutions (Eq. S2).

with linewidth  $\ell$  and expected drift  $d$  (defaults  $\sigma_{F2} = 0.03$ ,  $\sigma_{F1} = 0.30$  ppm for  $^1\text{H}-^{15}\text{N}$ ). The 1-D kernel, with  $s = \sigma/\Delta$  the width in pixels and radius  $r = \lceil 3s \rceil$ , is

$$g_{\sigma}(m) = \frac{\exp(-m^2/2s^2)}{\sum_{m'=-r}^r \exp(-m'^2/2s^2)}, \quad m = -r, \dots, r, \quad (\text{S4})$$

and the blurred (“lineshape-rendered”) image is the separable convolution

$$G_a = g_{\sigma_{F1}} *_{F1} (g_{\sigma_{F2}} *_{F2} R_a). \quad (\text{S5})$$

The continuous Gaussian is precisely what replaces the bin method’s *hard* bin edges with a *soft* response (see §S6).

#### S4 Mean-centring and the similarity score

Each image is centred by subtracting its global mean

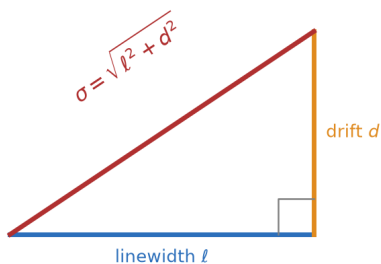
$$\bar{G}_a = \frac{1}{KL} \sum_{k,l} G_a[k,l], \quad \widehat{G}_a[k,l] = G_a[k,l] - \bar{G}_a, \quad (\text{S6})$$

and the similarity is the mean-centred normalized cross-correlation at **zero lag** (the Pearson correlation coefficient of the two flattened images), clamped to  $[0, 1]$ :

$$S(x, y) = \max \left( 0, \frac{\sum_{k,l} \widehat{G}_x[k,l] \widehat{G}_y[k,l]}{\sqrt{\sum_{k,l} \widehat{G}_x[k,l]^2} \sqrt{\sum_{k,l} \widehat{G}_y[k,l]^2}} \right) \quad (\text{S7})$$

Lineshape blur  $\sigma$  = physical linewidth and expected drift added in quadrature

(a) Blur width combines two physical widths



(b) The blur sets the shift-tolerance scale

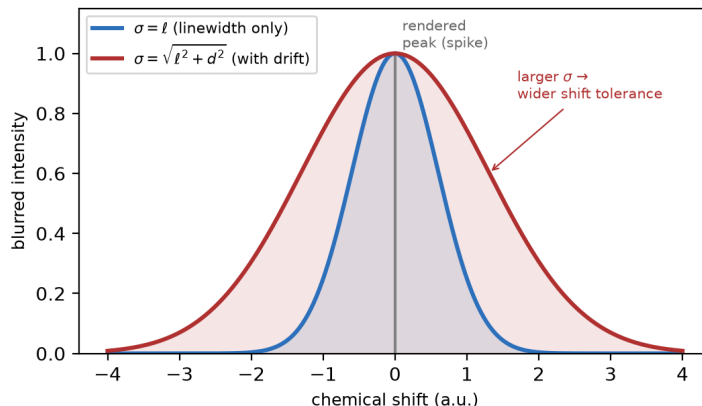


Figure 3: **Figure S3.** The lineshape blur width. (a) Each axis blur  $\sigma$  combines the physical NMR linewidth  $\ell$  and the expected chemical-shift drift  $d$  in **quadrature**,  $\sigma = \sqrt{\ell^2 + d^2}$  (Eq. S3). (b) Convolution of a rendered peak (a spike) with this Gaussian sets the shift-tolerance scale: a larger  $\sigma$  (linewidth *plus* drift) tolerates a larger displacement than the linewidth alone would.

No shift search / alignment is performed — this is deliberate (§S6c).

## S5 Properties

**Self-similarity is exactly 1.** For  $y = x$  the numerator and each factor of the denominator are the same sum  $\|\widehat{G}_x\|^2$ , so

$$S(x, x) = \frac{\|\widehat{G}_x\|^2}{\sqrt{\|\widehat{G}_x\|^2} \sqrt{\|\widehat{G}_x\|^2}} = 1, \quad (\text{S8})$$

and the  $\min(1, \cdot)$  clamp in the code absorbs the last-ULP floating-point wobble so an identical spectrum returns 1.0 bit-exactly.

**Symmetry.**  $S(x, y) = S(y, x)$  exactly, since the score is a symmetric inner product of the two centred images.

**Range.** As a Pearson coefficient  $S \in [-1, 1]$  before clamping; the clamp maps anticorrelation to 0, giving  $S \in [0, 1]$ .

**Monotonicity in drift.** For an isolated peak on a grid large enough that mean-centring is negligible,  $S$  is a strictly decreasing function of the displacement magnitude (Eq. S9); this is checked numerically in the test suite.

**Degenerate guard.** A flat image ( $\widehat{G}_a \equiv 0$ ) has zero denominator and returns 0; the windowing step already excludes empty windows upstream.

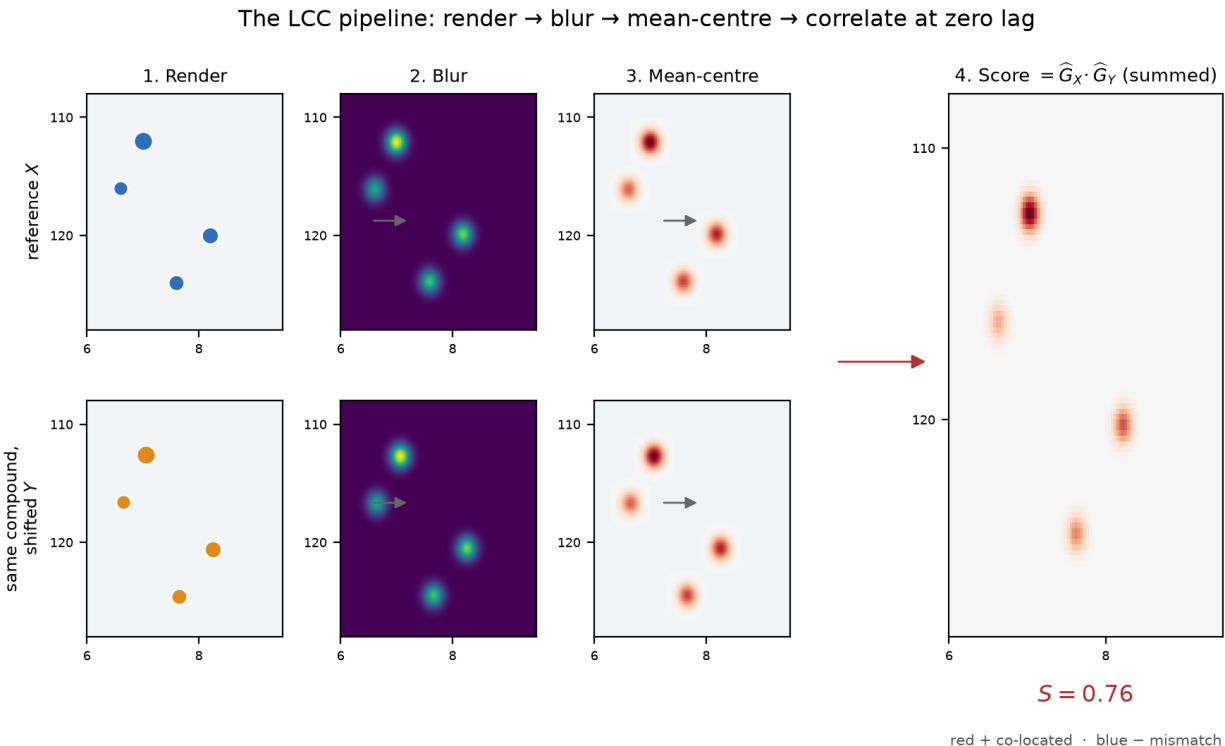


Figure 4: **Figure S4.** The four LCC steps end to end, on a toy pair. A reference  $X$  and a same-compound copy  $Y$  with a small titration drift are each (1) rendered as area-weighted points on a single **shared grid**, (2) blurred by the physical linewidth (a separable Gaussian), and (3) mean-centred (red = above the image mean, blue = below). (4) The similarity is the sum of the elementwise product  $\widehat{G}_X \cdot \widehat{G}_Y$  (normalized): co-located intensity gives a **positive** (red) product and drives the score up, so this same-compound pair scores  $S = 0.76$ . No shift search is performed — the images are correlated at zero lag.

## S6 Why LCC beats the existing methods

The three source methods fail in *opposite* ways: the bin method [1] is **shift-brittle** (a peak straddling a bin edge splits its mass, so a small titration drift drops the score in a discontinuous jump), while the tree [6,8] and nearest-neighbour (NN) [7] methods are **shift-blind** (in a dense fingerprint every peak has a near neighbour, so both saturate near 1). LCC sits between them via three mechanisms.

**(a) Graded shift tolerance.** Model a peak as a point mass; after the blur it is a Gaussian of width  $\sigma$ . Two copies of one peak displaced by  $(\Delta_{F2}, \Delta_{F1})$  contribute to the centred, normalized correlation a factor

$$\exp\left(-\frac{\Delta_{F2}^2}{4\sigma_{F2}^2} - \frac{\Delta_{F1}^2}{4\sigma_{F1}^2}\right), \quad (\text{S9})$$

which is **smooth and monotone** in the drift — no bin-edge discontinuity. A physical titration drift ( $\Delta_{1H} \sim 0.01\text{--}0.05$ ,  $\Delta_{15N} \sim 0.2\text{--}0.8$  ppm) is a fraction of  $\sigma$  and costs little; a random relocation ( $\Delta \gg \sigma$ ) drives the factor toward 0. The single knob  $\sigma$  sets the tolerance scale (main-text Figure 1b).

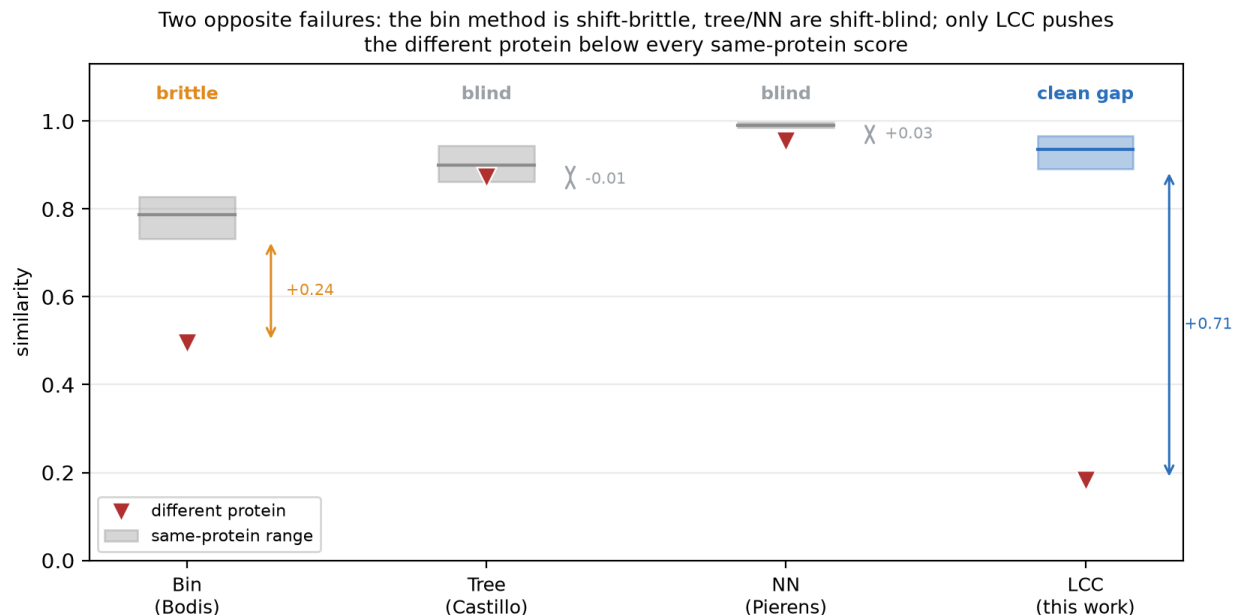


Figure 5: **Figure S5.** The two opposite failure modes, on the dense  $^1\text{H}$ - $^{15}\text{N}$  benchmark (Table S2). For each method the shaded **band** spans the same-protein similarities (the titration series) and the red marker is the **different protein**. The bin method is shift-**brittle** — a physical titration drift erodes its same-protein scores down to 0.73–0.83, the lowest band shown. The tree and NN methods are shift-**blind** — the different protein sits *inside* (tree) or right at (NN) the same-protein band, so it is not separated. Only LCC pushes the different protein (0.18) below *every* same-protein score, opening a wide clean gap.

Graded shift tolerance: LCC decays smoothly where the bin method jumps

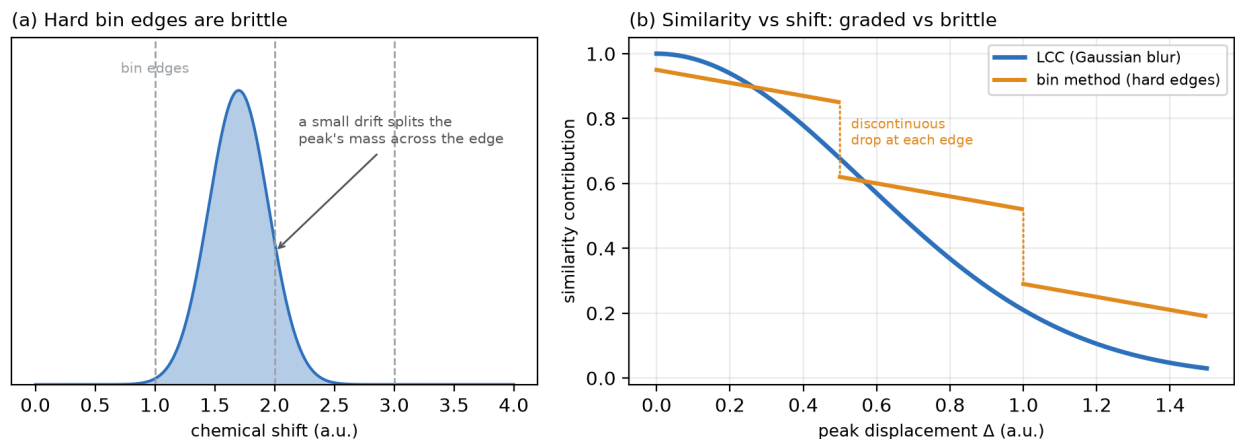


Figure 6: **Figure S6.** Graded vs brittle response to a shift. **(a)** The bin method’s edges are hard: as a peak drifts, a small displacement splits its mass across a bin boundary. **(b)** Similarity contribution vs peak displacement  $\Delta$ . The Gaussian blur makes LCC decay **smoothly and monotonically** ( $\exp(-\Delta^2/4\sigma^2)$ , blue), so a physical drift costs little; the bin method (orange) instead **jumps discontinuously** each time the peak crosses an edge — a small drift can cause a sudden score drop. This is why the bin method is shift-brittle and LCC is not.

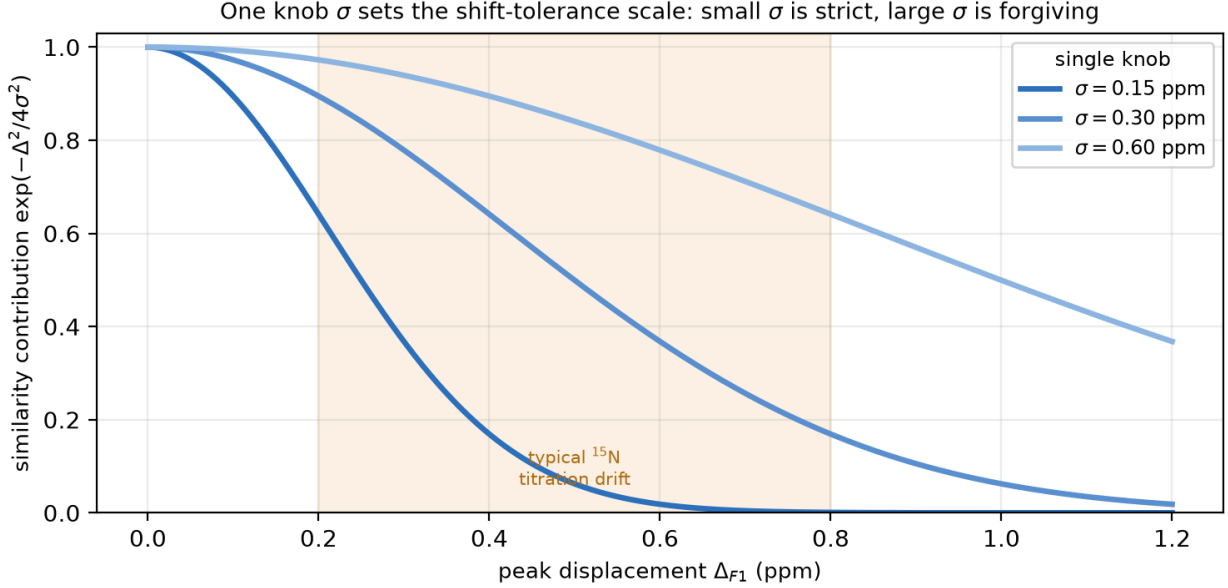


Figure 7: **Figure S7.** The single tuning knob. The same-peak contribution  $\exp(-\Delta^2/4\sigma^2)$  (Eq. S9) versus peak displacement  $\Delta$ , for three blur widths  $\sigma$ . One number sets the entire tolerance scale: a small  $\sigma$  is strict (a modest drift already costs similarity), a large  $\sigma$  is forgiving. The default is chosen so a typical  $^{15}\text{N}$  titration drift (shaded) costs little while a random relocation drives the contribution toward zero.

(b) **Coverage penalty from mean-centring.** Writing the score as a covariance,

$$S \propto \text{cov}(G_x, G_y) = \frac{1}{KL} \sum_{k,l} (G_x[k, l] - \bar{G}_x)(G_y[k, l] - \bar{G}_y), \quad (\text{S10})$$

a cell where one spectrum has a peak (value above its mean) and the other is empty (value below its mean) contributes a **negative** product and *lowers*  $S$ . Intensity that is not co-located is therefore actively penalized — a differently-scattered protein cannot hide behind coincidental proximity the way it does under NN’s one-way nearest-neighbour distance. This mean-centring step is worth  $\approx +0.16$  of separation over an un-centred overlap on the protein benchmark (§S10).

(c) **Zero lag on purpose.** Adding a global shift search (FFT / block matching), as used in spectral-alignment tools [9], would let a different protein slide into registration and re-saturate — the same failure as tree/NN. Scoring at zero lag keeps the position information that carries the discrimination.

## S7 Parameters

**Table S1.** LCC parameters. The two blur widths are the only tuning levers, and both are set by spectroscopy: the linewidth sets the floor, the expected shift perturbation sets the tolerance.

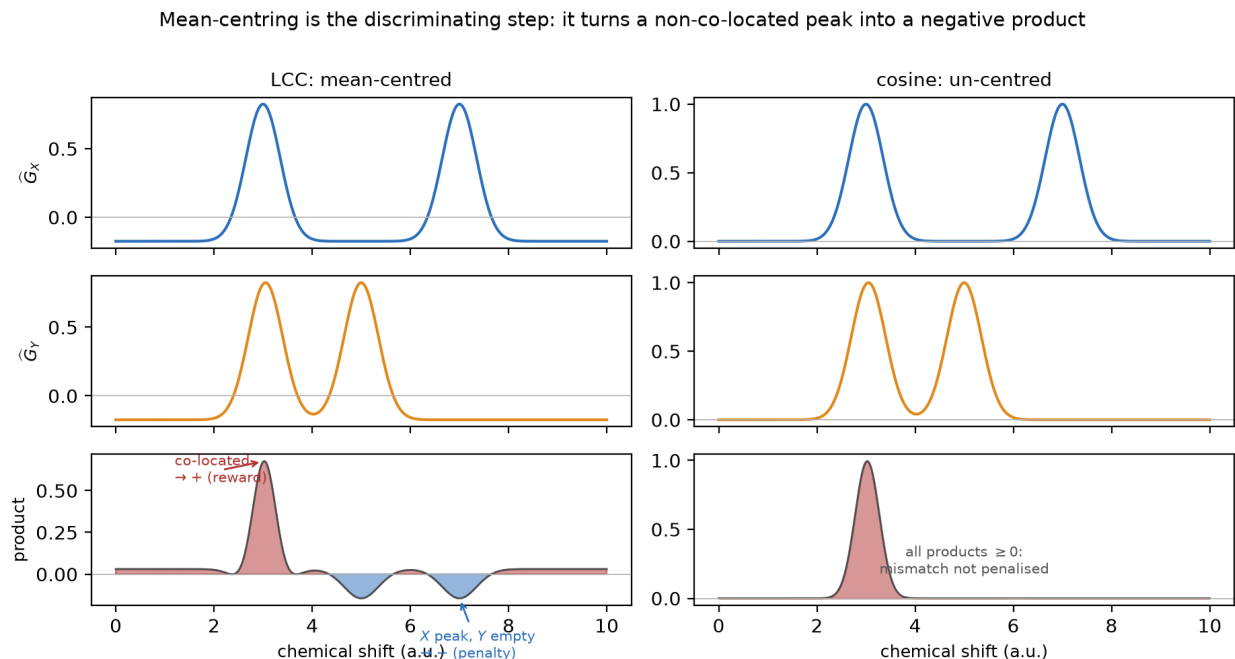


Figure 8: **Figure S8.** Why mean-centring is the discriminating step (1-D cross-section). Two images  $\hat{G}_X$  (blue) and  $\hat{G}_Y$  (orange) and their elementwise product (red = positive, blue = negative). **Left (LCC, mean-centred):** where both have a peak the product is **positive** (reward); where one has a peak and the other is empty, the empty image sits *below its mean* so the product is **negative** — non-co-located intensity is actively **penalized**. **Right (un-centred cosine):** every product is  $\geq 0$ , so a mismatched peak contributes nothing and cannot be penalized. Mean-centring is exactly what converts “overlap” into “discrimination”.

| symbol        | flag       | default ( $^1\text{H}$ – $^{15}\text{N}$ ) | meaning                                                                    |
|---------------|------------|--------------------------------------------|----------------------------------------------------------------------------|
| $\sigma_{F2}$ | --sigma-f2 | 0.03 ppm                                   | $^1\text{H}$ linewidth + drift blur width (main $F2$ tolerance)            |
| $\sigma_{F1}$ | --sigma-f1 | 0.30 ppm                                   | $^{15}\text{N}/^{13}\text{C}$ linewidth + drift (main tuning lever)        |
| $\Delta_2$    | --step-f2  | 0.01 ppm                                   | $F2$ render pixel width ( $\geq 3$ px per $\sigma$ ; score grid-invariant) |
| $\Delta_1$    | --step-f1  | 0.10 ppm                                   | $F1$ render pixel width                                                    |
| $\rho$        | --p        | 1.0                                        | intensity compression exponent (off)                                       |

Scoring at zero lag keeps position information; a global shift search would slide different spectra into false registration

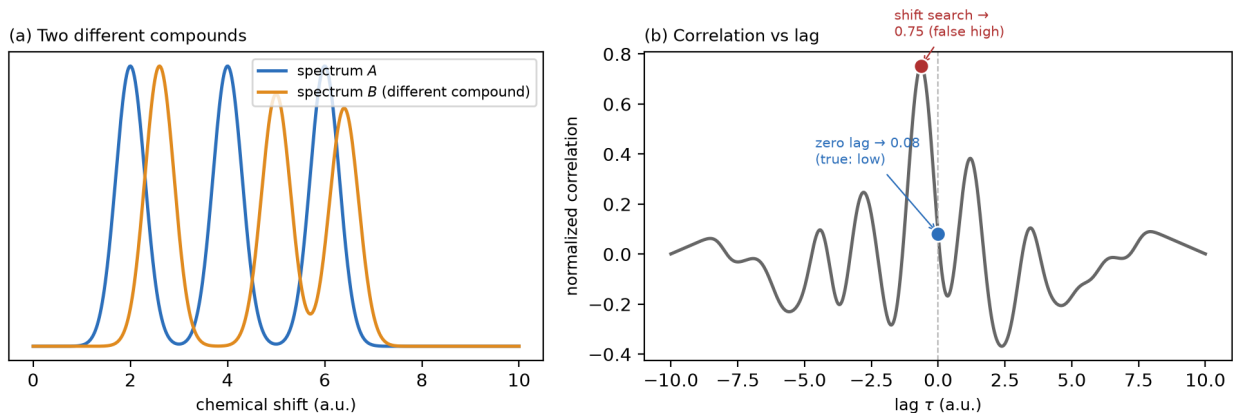


Figure 9: **Figure S9.** Why the score is read at **zero lag**. (a) Two genuinely different compounds (unequal peak spacing, so no rigid shift aligns them all). (b) Their normalized correlation as a function of a rigid lag  $\tau$ . At zero lag the score is correctly **low** (0.08, blue); a global shift search would instead report the **maximum** over all lags (0.75, red), sliding the different spectra into a partial false registration — the same over-tolerant saturation as tree/NN. LCC scores only at  $\tau = 0$ , keeping the position information that carries the discrimination.

Fixed by design: centring **on**, shift search **off**. Negatives are handled by the baseline mode (`clip` default; use `abs` for multiplicity-edited HSQC — §S1); `center=False` gives the un-centred cosine ablation (§S10). For the  $^1\text{H}$ - $^{13}\text{C}$  benchmark the blur widths are scaled to that regime ( $\sigma_{^1\text{H}} = 0.05$ ,  $\sigma_{^{13}\text{C}} = 0.5$  ppm).

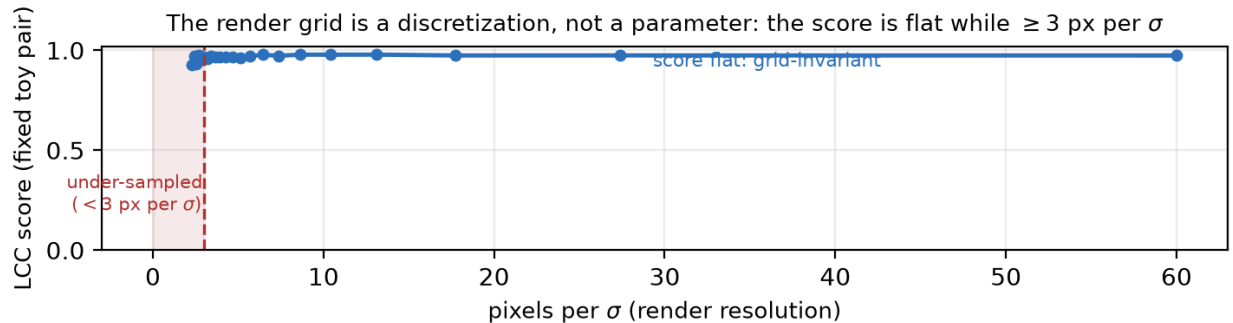


Figure 10: **Figure S10.** The render grid is a discretization, not a tuning parameter. A fixed same-compound toy pair is scored while the render pixel size is swept: the LCC score is **flat** while the Gaussian is sampled by  $\gtrsim 3$  pixels per  $\sigma$  (red zone = under-sampled). The choice of grid does not affect the result.

## S8 Benchmark 1 — dense protein $^1\text{H}$ - $^{15}\text{N}$

**Design.** Reference spectrum: PRL3 experiment 2, a  $^1\text{H}$ - $^{15}\text{N}$  HSQC. It is compared with the **same protein plus ligand** (experiments 4–14, a titration series that should score high — the chemical-shift-perturbation and fragment-based-screening setting [10–13]) and with a **different protein**

(OAA experiment 103, which should score low). This same-protein-versus-different-protein discrimination is the core operation of higher-order-structure (HOS) comparability assessment, for which 2D-HSQC fingerprinting is an established atomic-resolution readout (§S13). Window  $F_2$  6.5–10  $\times$   $F_1$  105–130 ppm. Bin widths for the bin method  $\text{min\_bin\_width\_f2} = 0.1$ ,  $\text{min\_bin\_width\_f1} = 1.0$ ; LCC blur  $\sigma_{F_2} = 0.03$ ,  $\sigma_{F_1} = 0.30$  ppm. Separation  $\Sigma = \bar{S}_{\text{same}} - S_{\text{diff}}$ ; margin =  $\min_e S_{\text{same}} - S_{\text{diff}}$ .

**Table S2.** Per-comparison similarity, dense protein  $^1\text{H}$ – $^{15}\text{N}$  benchmark. All methods self-score exactly 1.

| comparison                      | Bin (Bodis 2009) | Bin + 45° | Tree (Castillo 2013) | NN (Pierens 2012) | <b>LCC</b>    |
|---------------------------------|------------------|-----------|----------------------|-------------------|---------------|
| 2 vs 2 (self)                   | 1.0000           | 1.0000    | 1.0000               | 1.0000            | <b>1.0000</b> |
| 2 vs 4 (+B12)                   | 0.8106           | 0.8385    | 0.9401               | 0.9925            | <b>0.9662</b> |
| 2 vs 6 (+NB91)                  | 0.7739           | 0.7990    | 0.8619               | 0.9981            | <b>0.9594</b> |
| 2 vs 8 (+benz)                  | 0.8274           | 0.8570    | 0.9443               | 0.9877            | <b>0.9485</b> |
| 2 vs 10 (+GB001)                | 0.8164           | 0.8555    | 0.9179               | 0.9848            | <b>0.8972</b> |
| 2 vs 12 (+PF1)                  | 0.7637           | 0.7912    | 0.8760               | 0.9978            | <b>0.9558</b> |
| 2 vs 14 (+PF2)                  | 0.7320           | 0.7654    | 0.8624               | 0.9851            | <b>0.8911</b> |
| <b>2 vs OAA (diff. protein)</b> | 0.4949           | 0.5681    | 0.8721               | 0.9550            | <b>0.1815</b> |
| mean same – different (sep.)    | 0.29             | 0.25      | 0.03                 | 0.04              | <b>0.75</b>   |
| worst-case margin               | 0.24             | 0.20      | −0.01                | 0.03              | <b>0.71</b>   |

LCC separates same-protein from different-protein spectra  $\approx 2.6\times$  better than the previous best, pushing the different protein (0.18) below *every* same-protein score. The old bin method left the different protein (0.49) close to the worst same-protein point (0.73); LCC opens a wide gap.

**Expanded benchmark — a second decoy protein.** The benchmark above scores against a single decoy (OAA). To give the negative side a distribution over distinct folds — and to make the dense test self-contained — we added an independent PRL3 apo→olsalazine titration (500  $\mu\text{M}$  PRL3(4–159), 23 points from exp 4 to exp 48) and a second decoy protein, the kinase domain **EphB3**, all read as processed Bruker spectra in the same  $^1\text{H}$  6.5–10  $\times$   $^{15}\text{N}$  105–130 window

(`bench_nhsqc.py`). As the ligand is titrated the chemical-shift perturbation grows and LCC decays **smoothly** from 0.98 to a plateau near 0.82 (Table S9), the graded shift tolerance of §S6a acting on real titration data; both decoy proteins fall far below the entire titration (OAA 0.19, EphB3 0.41), so LCC’s margin (0.41) is positive across two distinct folds and the negative side is a distribution, not one spectrum. Mean-centring still earns the gap (LCC separation 0.57 vs un-centred cosine 0.46), and the tree and NN measures again saturate (margins 0.12, 0.02). The absolute margin is smaller than the six-point single-decoy benchmark (0.71) because this test is harder on both axes — a longer titration reaching larger shifts (min same 0.82 vs 0.89) and a tougher decoy (EphB3 0.41 vs OAA 0.19) — but the method ordering is unchanged.

**Table S8.** Expanded dense  $^1\text{H}$ – $^{15}\text{N}$  benchmark: PRL3 apo–olsalazine titration (23 same-protein points) vs two decoy proteins (OAA, EphB3). Separation =  $\bar{S}_{\text{same}} - \bar{S}_{\text{decoy}}$ ; margin =  $\min_e S_{\text{same}} - \max_d S_{\text{decoy}}$ . Every method self-scores 1.

| method                                              | mean same    | min same     | OAA          | EphB3        | separation   | margin       |
|-----------------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Bin<br>(Bodis<br>2009)                              | 0.817        | 0.753        | 0.461        | 0.500        | 0.337        | 0.253        |
| Bin +<br>45°                                        | 0.865        | 0.810        | 0.521        | 0.563        | 0.323        | 0.247        |
| Quad-<br>tree<br>(Castillo<br>2013)                 | 0.946        | 0.912        | 0.793        | 0.749        | 0.176        | 0.119        |
| Nearest<br>neigh-<br>bour<br>(Pierens<br>2012)      | 0.986        | 0.983        | 0.959        | 0.962        | 0.026        | 0.021        |
| Cosine,<br>un-<br>centred<br>(LCC<br>abla-<br>tion) | 0.897        | 0.853        | 0.360        | 0.507        | 0.464        | 0.346        |
| <b>LCC<br/>(this<br/>work)</b>                      | <b>0.871</b> | <b>0.820</b> | <b>0.195</b> | <b>0.414</b> | <b>0.567</b> | <b>0.405</b> |

**Table S9.** LCC along the PRL3–olsalazine titration (reference = apo exp 2): a smooth, near-monotone decay as ligand — and thus CSP — increases. Every titration point stays above both decoys (OAA 0.19, EphB3 0.41), so the worst same-protein score (0.82) still clears the worst decoy by 0.41.

| exp | 4    | 8    | 12   | 16   | 20   | 24   | 28   | 32   | 36   | 40   | 44   | 48   |
|-----|------|------|------|------|------|------|------|------|------|------|------|------|
| LCC | 0.98 | 0.95 | 0.93 | 0.90 | 0.87 | 0.85 | 0.85 | 0.84 | 0.83 | 0.83 | 0.83 | 0.82 |

## S9 Benchmark 2 — sparse small-molecule $^1\text{H}$ - $^{13}\text{C}$

**Design.** Public HSQC peak lists from the simpleNMR example set [14] (EricHughesABC/simpleNMR). Six nominal compounds, each recorded **twice** (a variant pair — e.g. menthol in two solvents). One nominal pair, **olivetol**, was excluded: its two source files (`Olivetol.json` and `Olivetol_A.json`) are **byte-identical** (same MD5, identical peak lists), so it was a self-comparison scoring 1.00 for every method — an information-free positive that inflated **mean\_same**. The benchmark now uses the **5 genuinely distinct** same-compound pairs (10 spectra) and asks each measure to score them above the **40 different-compound pairs**. `bench_13c.py` raises at load time on any same-pair that renders to an identical image, so a duplicate cannot silently re-enter. Peak lists are rasterized to a synthetic 2D spectrum (one Gaussian per peak, using  $|\text{intensity}|$  so edited-HSQC  $\text{CH}_2$  peaks are kept — §S1) so every method runs unchanged. Window  $^1\text{H}$  0–10  $\times$   $^{13}\text{C}$  0–165 ppm; LCC blur  $\sigma_{^1\text{H}} = 0.05$ ,  $\sigma_{^{13}\text{C}} = 0.5$  ppm.

**Table S3.** Aggregate scores, sparse small-molecule  $^1\text{H}$ - $^{13}\text{C}$  benchmark (5 same-compound pairs, 40 different). All methods self-score exactly 1. The un-centred cosine is LCC without mean-centring (§S6b), included here as the ablation baseline (bug-fix: it is now a reproducible benchmark row, not an asserted number).

| method                            | mean same    | mean diff    | separation   | margin       |
|-----------------------------------|--------------|--------------|--------------|--------------|
| Bin (Bodis 2009)                  | 0.712        | 0.067        | 0.646        | 0.195        |
| Bin + 45°                         | 0.740        | 0.091        | 0.650        | 0.181        |
| Quad-tree (Castillo 2013)         | 0.907        | 0.494        | 0.414        | 0.089        |
| Nearest neighbour (Pierens 2012)  | 0.997        | 0.896        | 0.101        | 0.043        |
| Cosine, un-centred (LCC ablation) | 0.781        | 0.004        | 0.777        | 0.370        |
| <b>LCC (this work)</b>            | <b>0.781</b> | <b>0.003</b> | <b>0.778</b> | <b>0.370</b> |

**Table S4.** Per-pair same-compound scores. Two pairs are genuinely shifted (menthol in two solvents; a rotenone re-measurement); on these the shift-tolerant tree/NN keep the same compound high while the position-sensitive bin/LCC penalize the shift. The catch is that tree/NN tolerate *everything*, which is why their different-compound scores (Table S3) are also high. The un-centred cosine column is identical to LCC to two decimals — in this sparse regime mean-centring changes nothing (see the ablation, §S10).

| pair                   | Bin  | Tree | NN   | Cosine | LCC  |
|------------------------|------|------|------|--------|------|
| menthol (two solvents) | 0.60 | 0.96 | 0.99 | 0.58   | 0.58 |
| rotenone (re-measured) | 0.30 | 0.83 | 1.00 | 0.39   | 0.39 |
| santonin               | 0.97 | 0.99 | 1.00 | 1.00   | 1.00 |
| chartreusin            | 0.80 | 0.83 | 1.00 | 0.94   | 0.94 |
| indanone               | 0.89 | 0.93 | 1.00 | 0.99   | 0.99 |

Over a large common window a different small molecule still supplies *some* peak near every peak and a similar mass-centre structure, so the NN distance and the tree overlap stay small — the same saturation mechanism as in the protein case.

## S10 Robustness and ablation

**Ablation — mean-centring is the discriminating step, in the dense regime.** Replacing the mean-centred correlation (Eq. S7) with an un-centred cosine overlap on the *same* rendered/blurred images drops the dense-protein separation from **0.75 to 0.59**, confirming mean-centring (§S6b) as the operation that turns overlap into discrimination when the amide fingerprint is crowded. In the **sparse**  $^1\text{H}$ - $^{13}\text{C}$  regime, by contrast, the un-centred cosine and centred LCC coincide (separation **0.777 vs 0.778**; Table S3, Table S4): where crosspeaks rarely overlap there is almost no non-co-located intensity for mean-centring to penalize, so its benefit is specific to the dense fingerprint that is the hard case this measure targets. The cosine baseline is a real, reproducible benchmark row in both harnesses (`cosine_similarity` in `hsqc_lcc.py`), not an asserted number.

**Robustness — the gain is not a tuned parameter.** On the protein benchmark LCC beats the bin method across the whole physical blur range: separation is  $\Sigma = 0.71$ – $0.77$  at  $\sigma = 0.02$ – $0.04$  /  $0.20$ – $0.40$  ppm, and still  $\Sigma = 0.36$  (vs the bin method’s  $0.29$ ) when the grid is coarsened all the way to the bin method’s own resolution ( $0.10$  /  $1.0$  ppm).

**Table S5.** LCC blur sweep, sparse  $^1\text{H}$ - $^{13}\text{C}$  benchmark — widening the blur recovers the shifted same-compound pairs while the different-compound mean stays near zero. This is the single physical knob that recovers the tree/NN shift tolerance without surrendering discrimination.

| $\sigma_{^1\text{H}} / \sigma_{^{13}\text{C}}$<br>(ppm) | mean same | min same | mean diff | separation |
|---------------------------------------------------------|-----------|----------|-----------|------------|
| 0.03 / 0.3                                              | 0.736     | 0.363    | 0.002     | 0.734      |
| 0.05 / 0.5                                              | 0.781     | 0.392    | 0.003     | 0.778      |
| 0.08 / 0.8                                              | 0.822     | 0.402    | 0.008     | 0.814      |
| 0.12 / 1.2                                              | 0.846     | 0.411    | 0.024     | 0.821      |

**Caveats.** (i) In the sparse benchmark, spectra are **rendered from peak lists**, not measured 2D matrices, so absolute values depend on the render width; the *ranking* is the result, not the exact numbers. (ii) The tree and NN methods here are the repository’s representative reimplementations run inside one common wide window; the original papers tune per-spectrum windows and parameters, so this is not a verdict on those methods as their authors deployed them — only on how they behave as drop-in global similarity scores. (iii) The between-compound contrast is strong (40 pairs) but the same-compound side is only 5 pairs, of which two (menthol, rotenone) are the genuinely-shifted, discriminating tests; a byte-identical sixth pair (olivetol) was excluded (§S9).

## S11 Relationship to the source methods

**Table S6.** LCC keeps the physically meaningful ingredient of each predecessor and adds the mean-centred zero-lag correlation as the new, discriminating step.

| ingredient                                                                                        | taken from                                      |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------|
| window + negative handling ( <code>clip/abs</code> ) + area weighting; normalization to unit mass | Bodis et al. 2009 (bins)                        |
| physical blur width as shift tolerance<br>( $\sigma = \sqrt{\ell^2 + d^2}$ )                      | Castillo et al. 2013 (tree shift-insensitivity) |
| a peak / lineshape picture of the spectrum                                                        | Pierens et al. 2012 (NN peaks)                  |

| ingredient                                                  | taken from |
|-------------------------------------------------------------|------------|
| mean-centred zero-lag correlation (the discriminating step) | this work  |

The un-centred cosine of the same rendered/blurred images (`center=False`) is the natural ablation of the last row — it isolates what mean-centring buys (§S10) and is the contrast-angle similarity of mass-spectral library search [15,16].

## S12 Software, reproduction and test suite

**Implementation.** LCC is pure NumPy [17] (`hsqc_lcc.py`); it reads the processed Bruker data (the `2rr` matrix and its `procs` parameters) directly through the repository’s own reader — without a general-purpose package such as `nmrglue` [19] — needs no peak picking and no external solver, and runs in about 6 ms per spectrum pair on a  $250 \times 350$  grid (Apple Silicon, single core). The processed input can come from any standard NMR processing pipeline [20–23]. Figures use Matplotlib [18].

**Reproduction.** Both benchmark harnesses are in the repository:

- `python3.11 bench.py` — dense protein  $^1\text{H}$ – $^{15}\text{N}$  benchmark (Table S2), writing `results/method_comparison.{csv,json}` and `results/lcc_comparison.png`.
- `python3.11 bench_nhsqc.py` — expanded dense benchmark with two decoy proteins (Tables S8, S9; main-text Table 2), reading the `Nhsqc/` PRL3–olsalazine titration and `OAA/EphB3` spectra and writing `results/nhsqc_dense.json`.
- `python3.11 bench_13c.py` — sparse small-molecule  $^1\text{H}$ – $^{13}\text{C}$  benchmark (Tables S3, S4), downloading the `simpleNMR` peak lists on first run and writing `results/comparison_13c.json`. It excludes the byte-identical olivetol pair and raises on any same-pair that renders identically.
- `python3.11 bench_retrieval.py` — dereplication/retrieval metrics (top-1, MRR) and bootstrap 95% CIs on the sparse regime (Table S7), writing `results/retrieval_13c.json`.
- `python3.11 results/make_plots.py` — regenerates Figure 2 (and `comparison_13c.png`) from the JSON; `python3.11 results/make_fig3.py` — regenerates Figure 3 (bootstrap CIs + ablation) from `retrieval_13c.json`; `python3.11 results/make_schematic.py` — regenerates Figure 1; `python3.11 results/make_si_figs.py` — regenerates the ten SI method figures (S1–S10).

**Test suite.** A 26-test suite (all passing) checks the properties proved in §S5 (self-similarity is exactly 1, related spectra score above unrelated ones, monotonicity in drift) plus the three bug-fix regressions: the un-centred cosine scores an unrelated pair *higher* than mean-centred LCC (so mean-centring is the discriminating step), the `abs` baseline preserves negative  $\text{CH}_2$  peaks that `clip` deletes, and a byte-identical same-pair is rejected. Run with `python3.11 -m pytest` (files under `tests/`: `test_hsqc_lcc.py`, `test_hsqc_methods.py`, `test_hsqc_similarity.py`, `test_bench_13c.py`, `test_spectrum_similarity.py`).

## S13 Statistical strength: bootstrap CIs and a retrieval test

Both additions run on the public sparse  $^1\text{H}$ – $^{13}\text{C}$  data and are reproducible with `python3.11 bench_retrieval.py` (writes `results/retrieval_13c.json`).

**Bootstrap confidence intervals.** The sparse-regime separation is resampled  $10^4$  times (independently resampling the 5 same-compound and 40 different-compound pairs) to give a non-parametric

95% CI [24] (Table S7). With only five same-compound pairs the CIs are wide: LCC’s interval overlaps the bin method’s, so **on this five-compound set LCC’s edge over the bin method is not statistically resolved** — both are strong. LCC’s interval is, however, cleanly above the saturating tree and nearest-neighbour intervals, so its advantage over the two shift-tolerant methods **is** significant. The honest reading is therefore: in the sparse regime LCC and the bin method are comparable and both beat tree/NN; LCC’s decisive, non-overlapping advantage is in the **dense** protein regime (margin 0.71). That margin was first measured against a single decoy; the expanded benchmark (§S8, Tables S8–S9) now scores an independent 23-point PRL3 titration against **two** decoy proteins (OAA and EphB3) and LCC keeps both below every same-protein point (margin 0.41), so the dense result no longer rests on one negative spectrum — while the sparse benchmark and this retrieval test broaden the evidence further with a larger public negative set.

**Table S7.** Sparse  $^1\text{H}$ – $^{13}\text{C}$  separation with a bootstrap 95% CI, and library-retrieval metrics. top-1 = fraction of the 10 query spectra whose rank-1 neighbour is the true same-compound partner; MRR = mean reciprocal rank [25] of that partner.

| method                            | separation   | 95% CI           | top-1       | MRR         |
|-----------------------------------|--------------|------------------|-------------|-------------|
| Bin (Bodis 2009)                  | 0.646        | 0.42–0.84        | 1.00        | 1.00        |
| Bin + 45°                         | 0.650        | 0.44–0.82        | 1.00        | 1.00        |
| Quad-tree (Castillo 2013)         | 0.414        | 0.34–0.49        | 1.00        | 1.00        |
| Nearest neighbour (Pierens 2012)  | 0.101        | 0.09–0.11        | 1.00        | 1.00        |
| Cosine, un-centred (LCC ablation) | 0.777        | 0.55–0.98        | 1.00        | 1.00        |
| <b>LCC (this work)</b>            | <b>0.778</b> | <b>0.55–0.98</b> | <b>1.00</b> | <b>1.00</b> |

**Retrieval (dereplication).** Cast as a library search — for each of the 10 spectra, rank the other 9 by similarity and ask whether the true same-compound partner is retrieved first — **every** method achieves top-1 = 1.00 and MRR = 1.00. This is an honest null result: on a small, clean library the correct partner is always the single most-similar spectrum, even for the saturating tree/NN methods whose *different*-compound scores are also high. Retrieval accuracy therefore does **not** discriminate the methods at this scale; the discriminating quantity is the thresholded same/different decision (separation and margin), which is what would matter against a large decoy library or with a rejection threshold. We report the null result rather than omit an experiment that did not favour the proposed method. A prospective, at-scale dereplication study [26,27] against a community library (BMRB [28], HMDB [29], NMRShiftDB [30]) is the natural way to turn this into a discriminating chemical application; LCC provides the scoring primitive, complementary to the multivariate workflows of NMR-based profiling [31]. A second, equally natural application is the **higher-order-structure (HOS) comparability** of protein biotherapeutics — biosimilar-

versus- originator, or before/after a formulation or process change — where the fold is confirmed by matching 2D-HSQC fingerprints and where 2D-NMR is an established, high-precision readout [32–37]; the dense-protein benchmark (§S8) is precisely this same/different discrimination task.

## S14 Supporting references

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